

EMAIL THREAT FORENSIC INVESTIGATION REPORT

OFFICIAL DIGITAL FORENSIC EXAMINATION DOSSIER // LAW ENFORCEMENT & SOC RECORD

Case Identifier	Examination Date	Investigation Status	Evidence Hash (SHA-256)
CASE-008	2026-09-03	VERIFIED EXAMINATION	3e6b9c0f4a7d1852c6e9b3a0f7d4...
Classification: COMPROMIS...	Risk Score: 74 / 100	Sample: HR_Updated_Payrol...	File Size: 128 KB

1.0 EXECUTIVE SUMMARY

A forensic examination was conducted on evidence sample 'HR_Updated_Payroll_Form.eml' cataloged under reference CASE-008. The electronic message claims transmission from sender 'Kavita Rao' (kavita.rao@iitd.ac.in). Automated and deterministic protocol evaluation assessed an authoritative risk score of 74 out of 100, resulting in a forensic threat classification of COMPROMISED_ACCOUNT.

Authentication protocol verification recorded SPF: PASS, DKIM: PASS, and DMARC: PASS. The message was accompanied by an identified lookalike typosquat target domain (office.com), indicating intentional brand impersonation. Routing telemetry analyzed 3 relay hops from origin to terminal destination server.

2.0 EMAIL & EVIDENCE SPECIFICATIONS

Evidence Field	Forensic Data Value	Verification Finding
Original Filename	HR_Updated_Payroll_Form.eml	Binary structure parsed and pres...
File Size	128 KB	Standard envelope size
Cryptographic Hash	3e6b9c0f4a7d1852c6e9b3a0f7d4c1e8a5b2f9d6c3e0a7f...	SHA-256 cryptographic integrity ...
Claimed Sender Name	Kavita Rao	Header-extracted display name
Claimed Sender Email	kavita.rao@iitd.ac.in	Header From address
Terminal Mail Server	New Delhi, India	Terminal destination host
Originating Node	14.162.88.203	First recorded external IP hop

3.0 AUTHENTICATION PROTOCOL ANALYSIS

Standard email authentication protocols were checked against published Domain Name System (DNS) records. Discrepancies indicate that the sending infrastructure was not authorized to transmit on behalf of the domain.

Protocol	Verdict	Technical Examination Finding
SPF (Sender Policy Framework)	PASS	Sending host verified against authorized SPF record
DKIM (DomainKeys Identified)	PASS	Valid cryptographic signature verified against DNS public key
DMARC (Domain Alignment)	PASS	Message satisfies alignment policy for domain

4.0 HEADER & ROUTING INFRASTRUCTURE

The message envelope traversed 3 recorded network transit hops. Network infrastructure was cross-referenced with autonomous system records and datacenter threat feeds.

Hop	IP Address	Geographic Location	Relay Timestamp	Forensic Infrastructure Note
#1	14.162.88.203	Hanoi, Vietnam	2026-09-06 03:22:11	Impossible-travel login origin
#2	103.27.9.18	New Delhi, India	2026-09-06 03:22:36	University authenticated relay
#3	103.27.9.20	New Delhi, India	2026-09-06 03:22:44	Destination mail server

5.0 LANGUAGE, INTENT & CONTENT ANALYSIS

Content linguistics and social engineering vectors were scored at 30 out of 30 maximum allocated points. Extracted textual phrases were analyzed for deception patterns, urgency cues, and financial solicitation.

Category / Signal	Detected Phrasing / Pattern	Confidence Assessment
CREDENTIAL REQUEST	"fill and submit the mandatory payroll direct deposit form"	92.0% Match
URGENCY	"failure to complete before payroll cutoff will delay salary"	88.0% Match

6.0 REPUTATION & DOMAIN INTELLIGENCE

Attribute	Observed Evidence	Analyst Interpretation
Domain Age	4 days	High anomaly: Domain registered immediately pr...
Typosquatting Target	office.com	Deliberate brand impersonation targeting legit...
Reputation Score Contribution	14 / 15 pts	Elevated risk: Associated infrastructure flagg...

7.0 INDICATORS OF COMPROMISE (IOCs)

The following actionable technical indicators were extracted during examination for perimeter blocklisting and SIEM correlation:

Indicator Type	Technical Value	Context / Threat Finding
IP Address (Origin)	14.162.88.203	Originating relay host observed in he...
Email Address	kavita.rao@iitd.ac.in	Header From address
Typosquat Target	office.com	Target brand impersonated in attack
Forensic Artifact	URL:hxxp://upd-payroll-cdn.net/form.hta	Extracted payment/URL artifact
File Hash (SHA-256)	3e6b9c0f4a7d1852c6e9b3a0f7d4c1e8a5b2f9d6c3e0a...	Unique cryptographic hash of message ...

8.0 ATTRIBUTION ASSESSMENT

ATTRIBUTION FINDING: LIKELY COMPROMISED LEGITIMATE ACCOUNT

Authentication protocols validated successfully; however, anomalous content indicates account takeover.
Note: Attribution conclusions represent technical forensic assessments based on available telemetry and do not constitute definitive legal identification without formal judicial subpoena records.

9.0 CORRELATION & INFRASTRUCTURE RELATIONSHIPS

Graph analysis cross-referenced entities across historical telemetry clusters and previous incident dossiers:

Source Entity	Target Entity	Correlation Basis / Linkage
email1	ip1	impossible travel origin
email2	ip1	same sending IP
email1	domain1	malware payload host

10.0 RISK ASSESSMENT & SIGNAL WEIGHTING

Overall Risk Score: 74 / 100 (Threshold evaluation: HIGH RISK). The score is derived across four standardized examination quadrants:

Quadrant Vector	Allocated Weight	Observed Score	Analytical Contribution
Header & Routing	30 pts	18 pts	Evaluates relay anomaly, ASN hosting, and h...
Authentication Failure	25 pts	12 pts	Evaluates SPF, DKIM signature validity, and...
Language & Intent	30 pts	30 pts	Evaluates urgency, financial diversion, and...
Reputation & History	15 pts	14 pts	Evaluates domain age, typosquatting targets...

11.0 KEY INVESTIGATION FINDINGS

- 1. High-confidence threat signature: Overall threat score exceeds enterprise containment threshold.
- 2. Authentication verified: Cryptographic signatures and domain alignment passed checks.
- 3. Lookalike brand impersonation identified: Attacker targeted legitimate entity 'office.com'.
- 4. Origin routing: Outbound transmission originated from standard commercial/workspace relay.

12.0 RECOMMENDED INVESTIGATION ACTIONS

- 1. Evidence Preservation: Retain full raw RFC 5322 message headers and message body in forensic storage.
- 2. Network Containment: Implement perimeter firewall block on originating IP 14.162.88.203.
- 3. DNS Mitigation: Issue internal DNS sinkhole rule for identified typosquatting target office.com.
- 4. Incident Response: Query mail gateway logs for all inbound messages matching the sending domain within past 30 days.
- 5. Account Safeguards: Enforce credential reset and session termination for any recipient who engaged with message content.

13.0 AUTOMATED NARRATIVE GENERATION ARCHITECTURE NOTE

SYSTEM NOTICE: This document was produced via deterministic evidence extraction from validated email forensic JSON. The

platform architecture is designed to integrate text-generation language models in future backend iterations to draft extended contextual DFIR narratives while maintaining the underlying JSON as the single source of truth.

14.0 EVIDENCE ATTESTATION & LEGAL DISCLAIMER

REPORT IDENTIFIER: REP-CASE-008 | PLATFORM: Email Threat Intelligence & Forensic Platform | EVIDENCE SOURCE: Ingested MIME/MSG Artifact | STATUS: VERIFIED FORENSIC RECORD

LEGAL DISCLAIMER: This document is an analytical examination record compiled from electronic message headers, cryptographic authentication tokens, DNS telemetry, and network routing hops. Findings should be corroborated with original raw headers, provider server logs, and judicial legal process before presentation in legal proceedings.